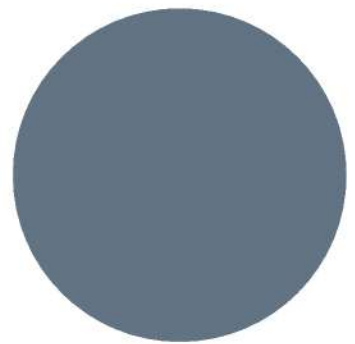

PUNE SMART CITY ENVIRONMENTAL SENSOR DATA ANALYSIS (2019)

Submitted by
Ganesh Babasaheb Gonge



ABOUT ME

- Background

I have completed a Bachelor of Science (BSc) in Information Technology from MIT College, Chh. Sambhiji Nagar, affiliated with Dr. Babasaheb Ambedkar Technological University (BAMU).

Why do you want to learn Data Science?

I want to learn Data Science because it enables me to turn large volumes of raw data into intelligent insights and predictive solutions that drive informed decision-making. I am deeply interested in how data can be used not only to analyze past performance but also to predict future trends and behaviors using machine learning and statistical models.

Data Science allows me to combine programming, mathematics, and domain knowledge to solve real-world problems such as optimization, forecasting, and automation. By learning Data Science, I will strengthen my skills in Python, SQL, data analysis, machine

learning, and data visualization, which will help me build a strong and future-oriented career in the Data Science and AI field.

LinkedIn and URLs:- <https://www.linkedin.com/in/ganesh-babasaheb-gonge-369a24277/>

GitHub profile URLs :- <https://github.com/ggonge16-star>



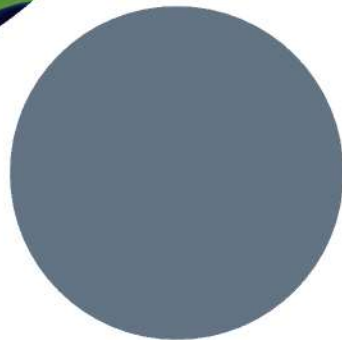
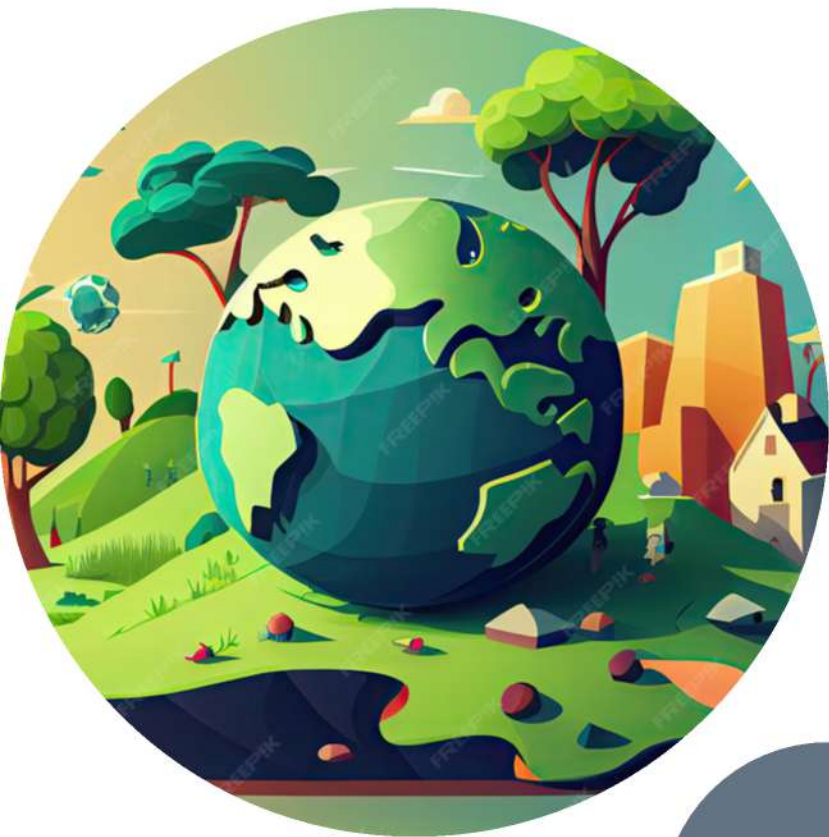
AGENDA

1. Introduction to Project
2. Objective of the Study
3. Dataset Overview
4. Data Preprocessing & Cleaning
5. Exploratory Data Analysis (EDA)
6. Outlier Detection
7. Data Visualization (Python / EDA)
8. Power BI Dashboard Overview
9. Key Insights & Findings
10. Conclusion
11. Future Scope



PROBLEM STATEMENT

Real-world datasets often contain missing values, outliers, and inconsistencies, which make it difficult to extract meaningful insights. Without proper Exploratory Data Analysis (EDA), identifying patterns, trends, and anomalies becomes challenging. The objective of this project is to analyze and clean the dataset using EDA techniques, detect outliers, and transform raw data into meaningful information. Furthermore, an interactive Power BI dashboard is developed to visualize key metrics, uncover hidden patterns, and support data-driven decision making.





DATASET OVERVIEW AND FEATURE CLASSIFICATION

Dataset: Pune Smart City Environmental Sensor Dataset (2019)

Records: 103,205

Features: 28

Data Type: Time-series + Location-based sensor data

Source: PSCDCL & IISc Bangalore

Location Data: Name, Latitude, Longitude

Air Pollutants: PM2.5, PM10, NO₂, CO, SO₂, Ozone

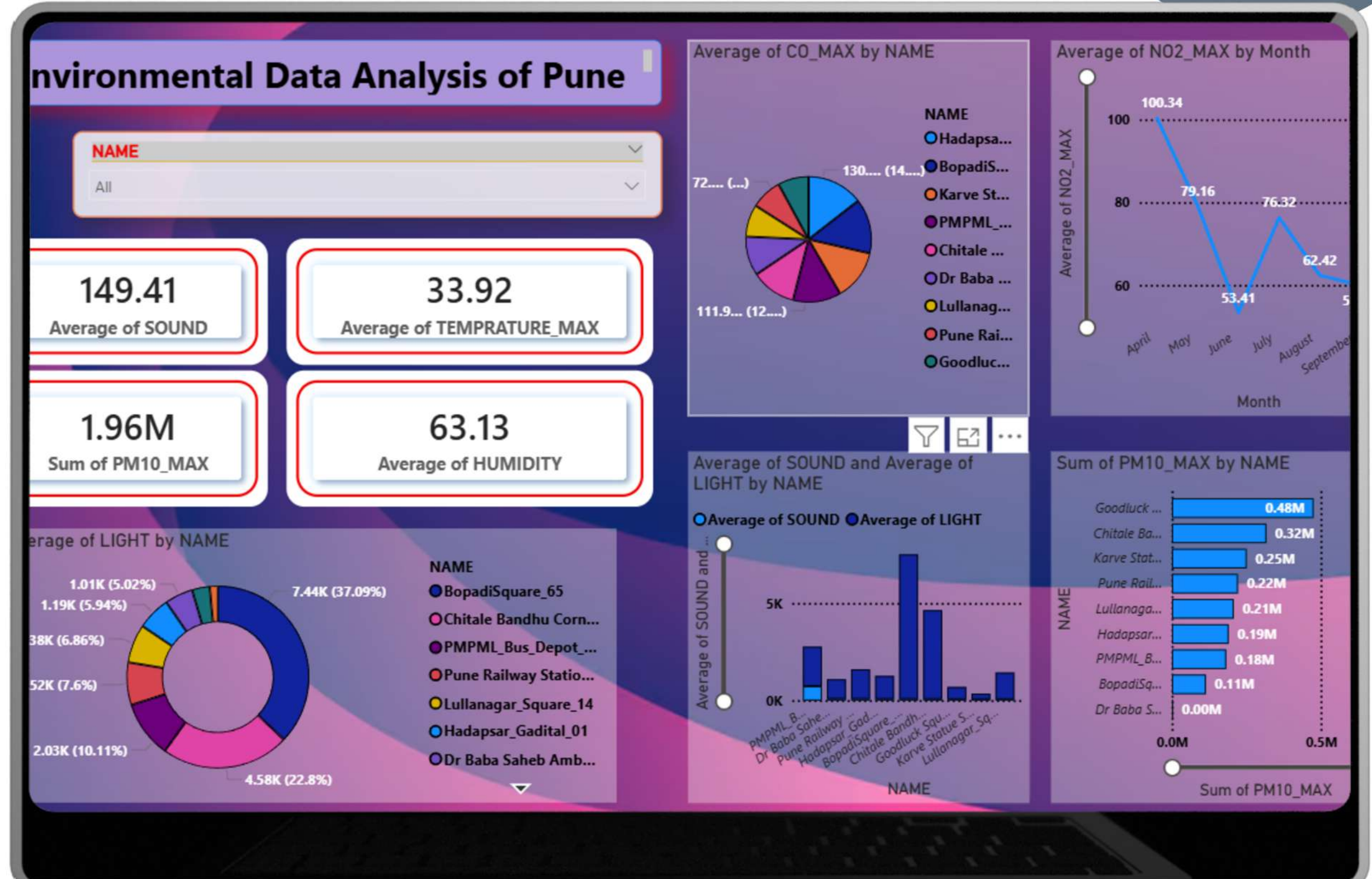
Environmental Parameters: Temperature, Humidity, Sound, Light, Air
Pressure

Time Feature: Timestamp, Hour, Day, Weekday

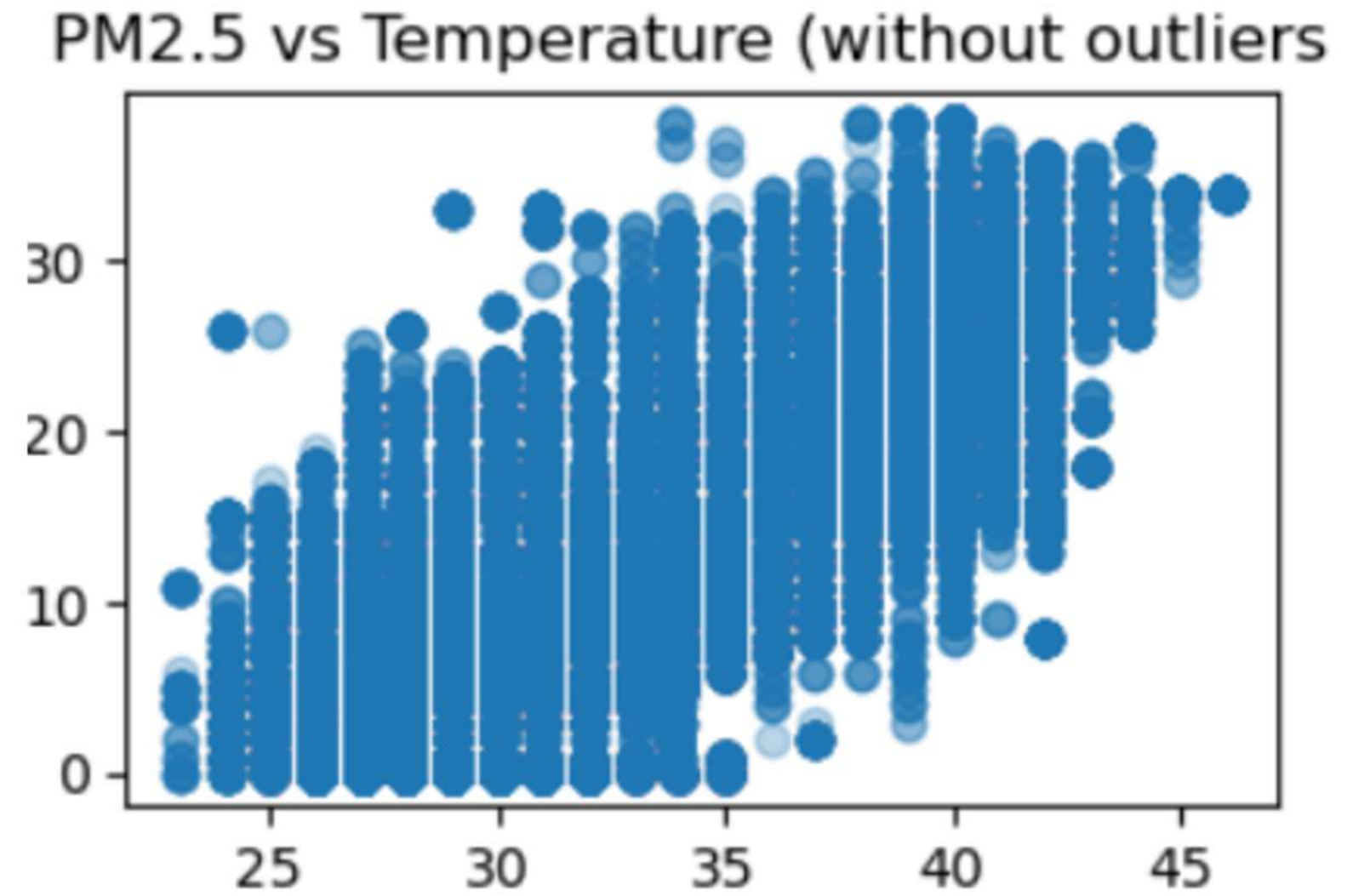
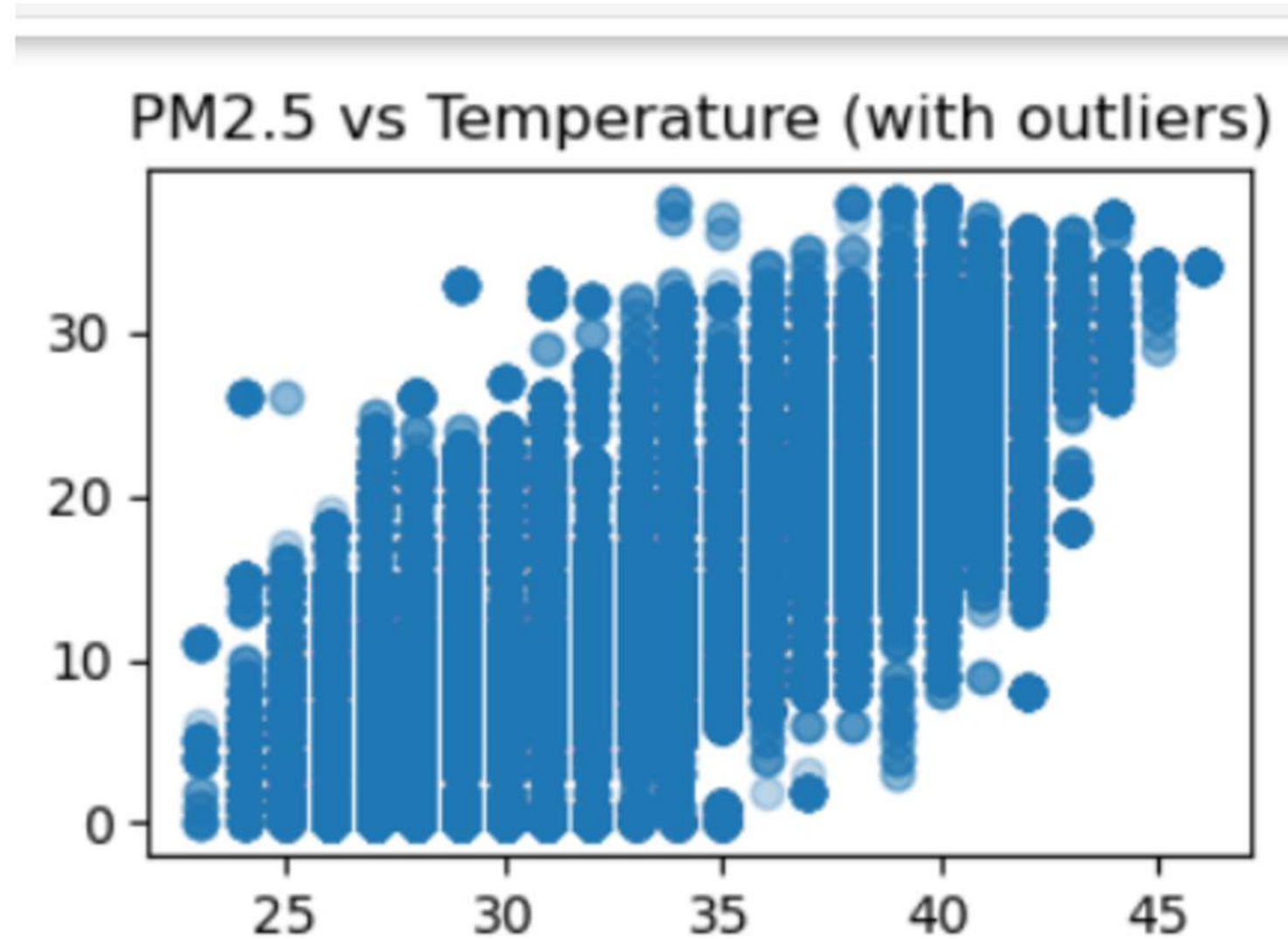


POWER BI DASHBOARD OVERVIEW

- Dashboard purpose
- KPIs shown
- Filters / Slicers used
- Interactive visualization



IS THERE A RELATIONSHIP BETWEEN TRAFFIC-RELATED POLLUTION (CO, NO₂) AND SOUND LEVELS?

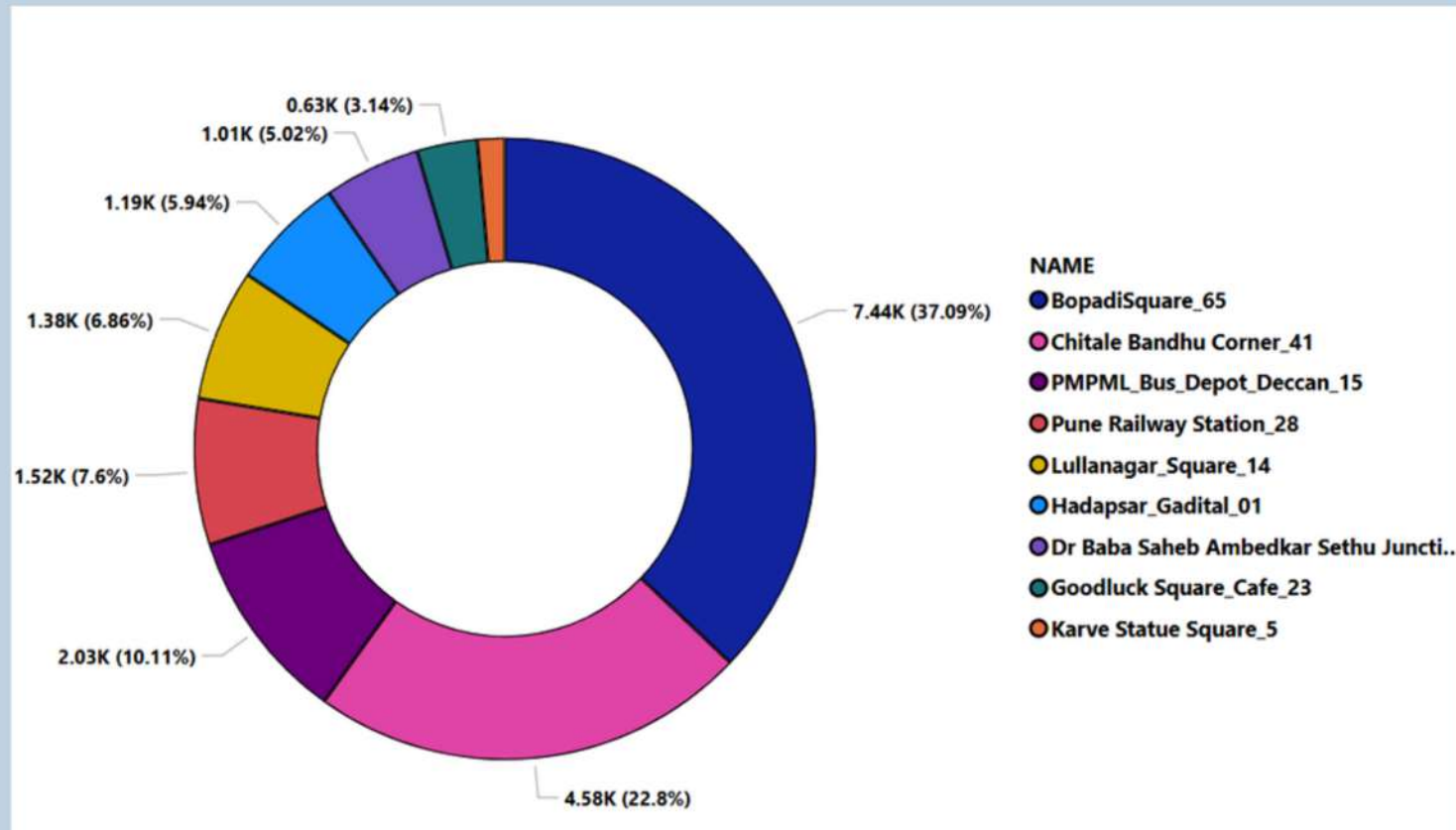


AVERAGE LIGHT INTENSITY BY LOCATION

This visualization shows the average light intensity recorded at different sensor locations in Pune city. The donut chart helps compare light levels across multiple monitoring points.

- Bopadi Square recorded the highest average light intensity, indicating higher brightness in that area.
- Chitale Bandhu Corner shows the second highest light level.
- Moderate light intensity is observed at PMPML Bus Depot (Deccan) and Pune Railway Station.
- Lower light intensity values are recorded at locations like Karve Statue Square and Goodluck Square, indicating relatively dimmer environmental conditions.

This analysis helps understand variation in light conditions across different areas, which may be influenced by traffic, urban activity, and infrastructure.



RESULT

The project successfully performed Exploratory Data Analysis (EDA) on the Pune Smart City Environmental Sensor Dataset and transformed raw sensor data into meaningful insights.

- Data cleaning and preprocessing improved overall data quality and consistency.
- Outlier detection helped identify abnormal environmental readings and reduced noise in the dataset.
- EDA revealed important patterns, trends, and relationships among environmental parameters such as air pollutants, temperature, humidity, and light.
- Time-series analysis showed variation of environmental conditions over different hours and days.
- The Power BI dashboard provided clear and interactive visualization of pollution levels, environmental parameters, and location-wise comparisons.
- Key environmental insights were identified, helping in better understanding of urban environmental conditions and monitoring.

Overall, the project demonstrated how EDA and data visualization can convert raw environmental data into actionable insights for data-driven decision making.

CONCLUSION

This project successfully applied Exploratory Data Analysis (EDA) techniques to clean, preprocess, and analyze the Pune Smart City Environmental Sensor Dataset. Data preprocessing improved data quality by handling missing values and reducing noise through outlier detection.

EDA helped in identifying important patterns, trends, and relationships among environmental parameters such as air pollutants, temperature, humidity, and light intensity. Time-based analysis provided a better understanding of how environmental conditions change over different hours and days.

The Power BI dashboard effectively visualized location-wise environmental data, pollution levels, and key metrics through interactive charts and filters, making the data easy to understand and interpret.

Overall, the project demonstrates the importance of data cleaning, EDA, and visualization in transforming raw environmental data into meaningful insights for environmental monitoring and data-driven decision making.

Thank You

